FallingDown AI

From Falling Down to Rising Strong.

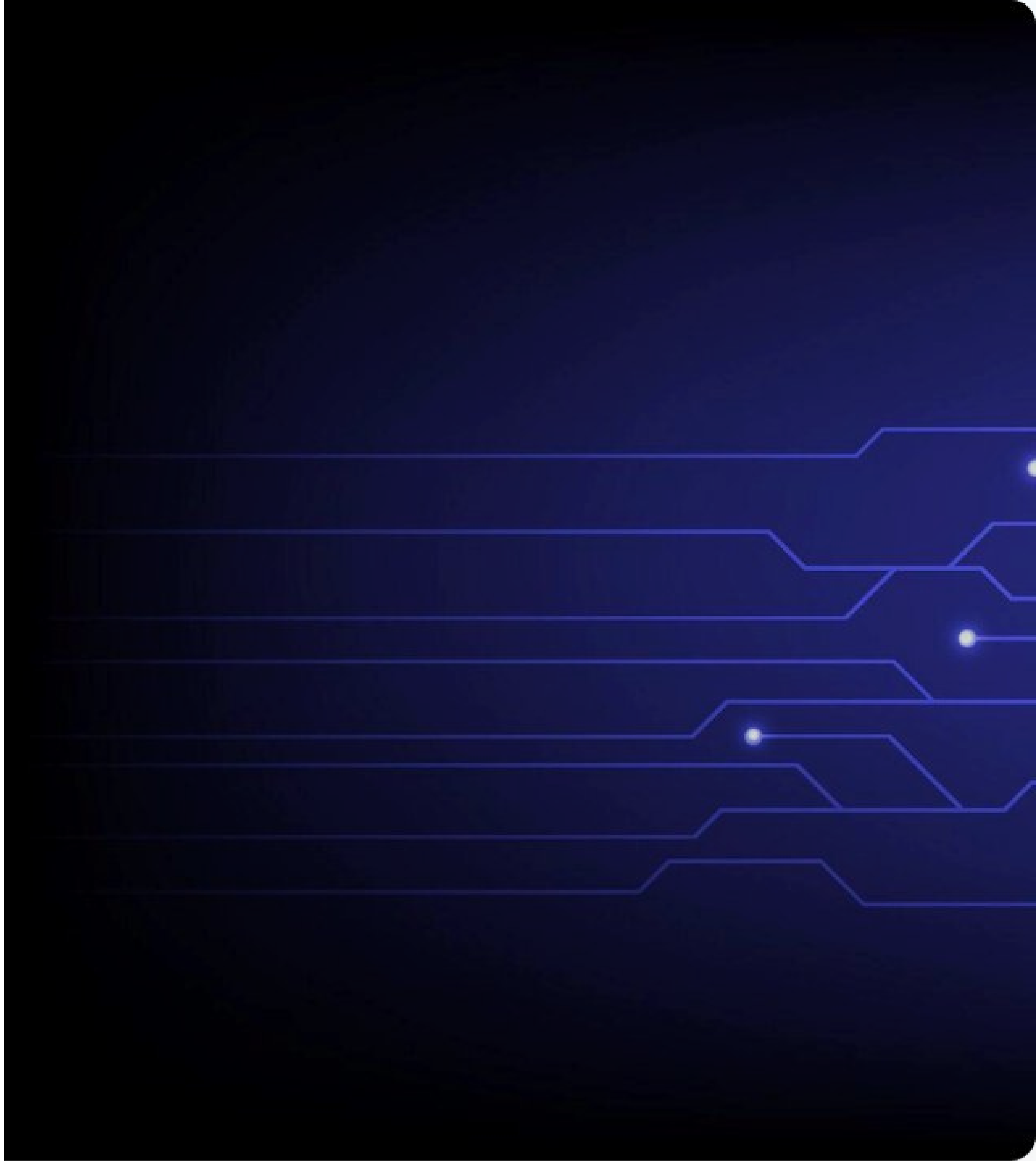
Intelligent Emergency Monitoring System

PRESENTED BY: THREE FALL

Avik Chakraborty • Barun Kumar Ghosh • Aniket Pandey

Haldia Institute of Technology

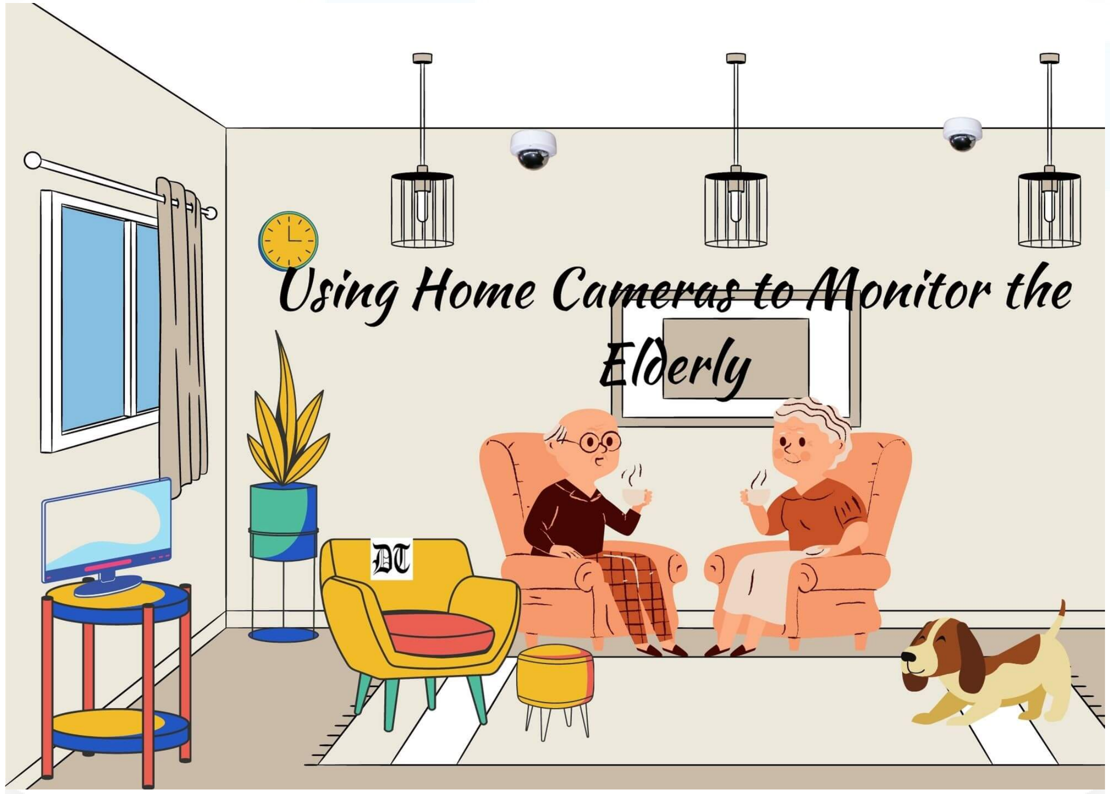
Haldia, West Bengal



The Problem: Silent Emergencies

The "Long Lie" can be fatal. Why are current systems failing our elderly?

- **Delayed Response:** Falls often remain unnoticed for hours when elderly individuals live alone, drastically increasing injury risk.
- **The Privacy Paradox:** The unique privacy paradox of elder care—ensuring safety without intrusive surveillance.
- **Passive Inefficiency:** Traditional CCTV systems only record the incident; they lack the intelligence to actively detect and alert families automatically.
- **Human Limitation:** Family members or nursing staff cannot physically monitor screens 24/7.



FallingDown AI: Proactive Intelligence

We transform ordinary smart cameras into intelligent emergency monitoring systems. Real-time AI detection combined with privacy-aware alert protocols.

1. AI Edge Monitoring

Advanced posture and motion analysis algorithms run locally on the edge. No sensitive video footage is ever uploaded to the cloud.

2. Privacy-Aware Zones

For sensitive areas like bathrooms, the system switches to "Audio Safety Mode," detecting distress vocalizations and loud impacts instead of video.

3. Instant Alert Pipeline

If a critical fall is detected, automated alerts (with redacted photo evidence) are sent instantly via WhatsApp to the designated Care-Circle.

Target Market Segments

B2C: Home & Family Care

- **Target Users:** Families with aging parents, elderly individuals living alone, and independent home caregivers.
- **Primary Pain Point:** Constant anxiety regarding the safety of loved ones when left unattended.
- **Buying Motivation:** 24/7 peace of mind, remote monitoring capability, and the assurance of instant emergency response.

B2B: Institutional Care

- **Target Users:** Hospitals, assisted living facilities, and dedicated nursing homes.
- **Primary Pain Point:** High patient-to-nurse ratios, inability to monitor all wards simultaneously, and liability risks.
- **Buying Motivation:** Automated patient monitoring, reduced nurse fatigue, and rapid emergency escalation.

Competitive Advantage

	FallingDown AI	Apple-watch	Standard CCTV	Panic Buttons
No Wearable Required	✓	✗	✓	✗
Automated Fall Detection	✓	Partial	✗	✗
Privacy-Aware (No Video Cloud)	✓	✓	✗	✓
Bathroom Audio Safety Mode	✓	✗	✗	✗
Instant WhatsApp / Photo Alert	✓	✗	✗	✗

Unique Selling Proposition (USP)

- **Zero-Friction Monitoring:** The elderly do not need to remember to wear, charge, or activate any device. The system works ambiently.
- **Advanced AI Understanding:** We don't just detect motion; the AI differentiates between a normal resting position, a dropped object, and a critical human fall.
- **Retrofit Capability:** Plugs into existing IP/Smart camera infrastructure, dramatically reducing setup costs for hospitals and homes.
- **Automated Evidence:** When a fall occurs, the system captures a privacy-blurred image and sends it via WhatsApp, allowing guardians to assess severity instantly.
- **Care-Circle Ecosystem:** Alerts escalate sequentially—from local siren, to primary caregiver phone, to secondary family members, ensuring someone always responds.

Go-To-Market Strategy

Home Market (B2C)

Digital Acquisition: Targeted social media campaigns focusing on NRIs and working professionals living away from their aging parents.

Channel Partners: Collaborating with local CCTV installers and smart-home integrators to offer FallingDown AI as a premium add-on.

Community Trust: Conducting free awareness seminars in elder-living communities and leveraging word-of-mouth referrals.

Institutional (B2B)

Pilot Programs: Offering 30-day risk-free pilot installations in high-fall-risk wards of targeted local hospitals to prove efficacy.

Strategic Partnerships: Partnering with established healthcare providers, insurance companies, and nursing home associations.

Direct Outreach: Dedicated B2B sales team conducting live demonstrations for hospital administrators focusing on ROI and liability reduction.

Monetization & Revenue Streams

A realistic, scalable business model focusing on high-margin SaaS and optional hardware bridging.

CORE REVENUE

B2C Home Subscription

₹2,999 – ₹49,999 / Month

- AI monitoring included
- Instant WhatsApp alerts
- Dashboard access
- Software updates
- Customer support

OPTIONAL

Hardware & Setup

*Only for users without existing compatible cameras.

₹3,599 – ₹4,599 (One-Time)

Setup Breakdown:

- 1 WiFi / IP Camera
- Optional Mic/Audio Layer
- Local Edge Setup
- Installation & Onboarding

B2B GROWTH

Nursing Home Plan

₹1,499 – ₹2,999 / Month

- 4–5 Camera Monitoring
- Central Admin Dashboard
- Multi-room Tracking
- Staff Alert Access
- WhatsApp Escalations
- Basic Deployment & Support

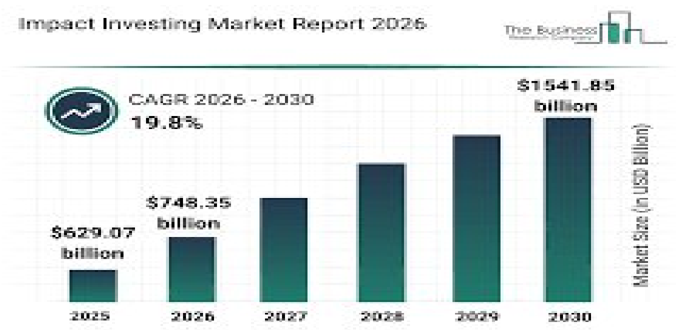
Unit Economics Comparison (Monthly SaaS Per Unit)

Parameter	B2C Model (Per Unit)	B2B Model (Nursing Home)	Description
Monthly SaaS Metrics			
Monthly Subscription	₹499	₹3,999	• Recurring SaaS fee
SMS/WhatsApp & API Cost	₹60	₹450	• User messaging & API access
Cloud/Server Maintenance	₹80	₹600	• Data processing & hosting
Support & Maintenance	₹100	₹800	• Customer service & platform updates
Net Monthly Profit	**₹259**	**₹2,149**	• Profit per month • Profitable monthly recurring
Operating Margin	**52%**	**54%**	• Recurring profit margin

This analysis focuses on recurring monthly SaaS metrics for both B2C (single home) and B2B (small nursing home, typically 8-10 units) models.

Hardware Unit Economics (similar to B2C day-one profitability) apply per device.
B2C Hardware Unit Economics (Day 0, per device): Selling Price: ₹4,999; Cost: ₹2,500; Gross Profit: ₹2,499.

Impact & Future Vision



Social Impact

Providing dignity, safety, and faster emergency response for the elderly, while drastically reducing caregiver stress and anxiety.

Economic Impact

Preventing long-term hospitalization costs through early intervention, and creating a new tech-support ecosystem jobs.

Eco & Future

Reusing existing hardware reduces e-waste. Future expansion aims to integrate full smart-home health ecosystems.

FallingDown AI: Because every second counts.